

# Hungi Jung

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Google Scholar   ORCID   OpenReview   LinkedIn

AI researcher focused on medical data modeling and clinical NLP, with hands-on experience in LLM-based EMR extraction, time-series modeling, and AWS-based deployment.

## Education

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**B.S. in Computer Science**, Soongsil University

*Mar 2022 – Feb 2025*

## Experience

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**Research Intern**, Center for Artificial Intelligence in Healthcare (CAIH) *Sep 2025 – Present*  
Seoul National University Bundang Hospital

- Developed an LLM-based hybrid model for gastric cancer EMR processing
- Designed a structured extraction pipeline for pathology registry variables

**Undergraduate Research Intern**, PEM Lab, Hongik University

*Feb 2024 – Feb 2025*

- Developed an AI model for arrhythmia detection using Python and Colab
- Contributed to signal preprocessing logic and model stabilization
- Presented the research at ACK 2024 (Korea Information Processing Society)

## Training

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**Biohealth AI Academy**, AWS × Sungkyunkwan University SAY

*Feb 2025 – Sep 2025*

- Participated in AWS official training on cloud computing and AI-based data analysis
- Conducted pre-project and final project using AWS cloud services for biohealth AI

## Publications

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**Structured LLM Pipeline for Automated Gastric Cancer Registry Construction from Bilingual Pathology Reports**

Hungi Jung, So Hyun Kang, Myeong ju Kim

*Under review, MLHC 2026 Clinical Abstract Track*

**Wavelet-Based ECG Feature Extraction and CNN for Arrhythmia Classification: An Enhanced Approach Using MIT-BIH Database**

YunSeo Jo, Hungi Jung, SeungJu Oh, HaYoon Song

*Annual Conference of KIPS, 2024*   [Link]

## Projects

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**Extraction and Structuring of EMR Data Using a Query-Regex-LLM Hybrid Model**  
*Sep 2025 – Present*

- Extracted structured clinical data from SNUBH CDW to build registry-ready datasets
- Designed a hybrid pipeline combining regular expressions and LLMs for EMR parsing
- Implemented data transformation logic to standardize heterogeneous clinical records
- Developed a scalable pipeline for automated registry data generation

## **ML-based Mortality Prediction Dashboard using MIMIC-IV** *Jul 2025 – Sep 2025*

- Extracted key features from MIMIC-IV to predict in-hospital mortality across wards, ICUs, and EDs
- Trained XGBoost, Transformer, and TCN models; selected optimal model by performance metrics
- Applied SHAP and LIME for model explainability
- Developed an AI-based dashboard for healthcare professionals

## **CNN + XAI-based Brain Tumor Detection** *Jun 2025 – Aug 2025*

- Utilized an open-access brain tumor dataset from Kaggle
- Designed preprocessing and classification tasks using CNN models
- Applied Grad-CAM for model interpretability

## **AI-based Arrhythmia Detection Using MIT-BIH Dataset** *Jun 2024 – Apr 2025*

- Designed a deep learning model for arrhythmia detection from ECG data
- Built a preprocessing pipeline incorporating cardiac domain knowledge and signal characteristics

## **Research Interests**

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Clinical NLP (EMR, LLM-based extraction) · Machine Learning & Deep Learning · Explainable AI (XAI)  
Time-series modeling (ECG, HRV) · Medical AI systems & deployment (AWS) · Computer Vision

## **Skills**

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**Languages:** Python, Java, SQL

**ML/DL:** PyTorch, scikit-learn, XGBoost, pandas, numpy, matplotlib, wfdb

**Cloud:** AWS

**Tools:** Git, Jupyter Notebook, Google Colab

## **Certifications**

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**Network Advisor Level 2, ICQA**

*Jan 2023*